

BAGOMBEKA JOB

Software Engineer | Full-Stack Delivery at National Scale | Laravel · Vue.js · Python · Java · PostgreSQL

Kampala, Uganda | Open to 100% Remote (EMEA / GMT+3) | +256 778 480 981 | bagombekajob16@gmail.com

linkedin.com/in/bagombeka-job | github.com/BAGOMBEKA-JOB-DEV | bagombekajob.com

Contract-ready via Deel, Oyster or Payoneer — no relocation required

PROFESSIONAL SUMMARY

Software engineer with over 5 years of production experience building scalable, user-facing systems, including national government platforms that have registered more than 30 million users and handle 45M+ records. Full-stack across PHP/Laravel, Vue.js, Python and Java, with a strong grounding in REST API design, relational data modelling and system architecture at scale.

Owns modules end to end — requirements, data model, SRS, development, testing, release and post-deployment support — and has authored the 1,000+ table PostgreSQL schema, data models and system architecture that became the reference specification for a national delivery team. Has led small cross-functional teams to a 40% improvement in delivery, driven a 70%+ reduction in recurring production defects through systematic root-cause analysis, and sustained 99.9% database uptime on business-critical systems. Comfortable integrating with hostile, undocumented third-party systems, and committed to clean, maintainable, well-documented code on mission-driven software.

CAREER HIGHLIGHTS

- 30M+ learners registered nationwide through the national learners module built in Laravel and Vue.js, with identity verification at the point of ingest eliminating duplicate registrations.
- 1,000+ table PostgreSQL schema designed and documented, together with the data models, SRS and system architecture adopted as the reference specification by the delivery team.
- 45M+ records handled on the Parliament of Uganda and State House messaging platform, with SMPP integration and structured high-volume ingestion.
- 100,000+ concurrent users supported with minimal downtime on a high-availability architecture designed and built from scratch.
- 70%+ reduction in recurring production defects achieved by replacing surface patches with systematic root-cause analysis.
- 40% improvement in internal workflows and delivery achieved while leading a cross-functional engineering team through planning, code review and release.
- 99.9% uptime sustained on SQL database servers through disciplined installation, configuration, monitoring and maintenance.
- Five external institutions integrated — national identity (NIRA), examinations (UNEB), teacher management (TMIS), the Office of the Prime Minister and the Ministry of Finance — across agencies that had never shared data before.

TECHNICAL SKILLS

Languages: PHP, JavaScript (ES6+), Python, Java, SQL, PL/SQL, Go (Golang), C++, Dart, Bash, HTML5, CSS3.

Back-End & Frameworks: Laravel (Eloquent ORM, Blade, Livewire, Filament, Artisan, Queues, Jobs, Events, Migrations, Seeders, Composer, Policies, Middleware), Python (FastAPI, Flask, Django, Pandas), Java (Spring Boot), Go, MVC architecture, service and repository patterns, domain modelling, background job processing, scheduled tasks.

Front-End: Vue.js (components, composition API, state management, routing), Alpine.js, React.js, JQuery, Node.js, Tailwind CSS, Bootstrap, Blade templating, responsive and accessible design, complex data-entry forms, data-heavy dashboards and tables, Flutter for cross-platform mobile.

Databases & Data Modelling: PostgreSQL, MySQL, MongoDB, PL/SQL, SQL Server; relational data modelling and normalisation, large-schema design (1,000+ tables), constraints and referential integrity, indexing strategy, query optimisation and execution-

plan analysis, migrations and versioned schema change, partitioning and archiving considerations, bulk ingestion, deduplication and reconciliation, backup and recovery.

APIs & Integration: REST API design and versioning, JSON, OAuth 2.0 and token-based authentication, Python FastAPI services, web services and third-party integration with national identity, workforce, examinations and finance systems, SMPP and Kannel messaging gateways, webhook and callback handling, idempotency, retry and reconciliation strategies, API documentation.

Architecture & Design: System architecture and high-level design, modular monolith and microservices, service boundaries, high-availability and scalability design, caching and queueing, event-driven patterns with RabbitMQ and Kafka, object-oriented analysis and design (OOAD), design patterns, software requirements specification (SRS), technical specification and decision documentation.

DevOps, Tooling & Observability: Git and Git-flow, Docker and containerised environments, CI/CD pipelines, Linux (Ubuntu/CentOS) administration and shell scripting, Nginx/Apache, environment and configuration management, release and rollback planning, Elastic Stack (Elasticsearch, Logstash) for logging and search, monitoring, alerting and log analysis, AWS and Azure fundamentals, Firebase.

Quality, Testing & Security: Unit, integration, regression and user acceptance testing, QA process support, code review, static analysis, debugging and systematic root-cause analysis, defect triage, secure coding, input validation, authentication and role-based authorisation, least-privilege design, audit logging, vulnerability assessment and penetration testing (Burp Suite, OWASP ZAP, Nmap, Metasploit).

Practices & Collaboration: Agile/Scrum, sprint planning, estimation, backlog refinement, JIRA, technical documentation, mentoring and code review culture, stakeholder communication and requirements elicitation, cross-functional team leadership, remote and asynchronous collaboration across time zones.

PROFESSIONAL EXPERIENCE

SMSONE Uganda Ltd — Kampala, Uganda | smsone.co.ug
Software Engineer

Jan 2025 – Aug 2026

Key platforms: EMIS — Education Management Information System (Ministry of Education and Sports, Government of Uganda) and the BulkSMS Platform (Parliament of Uganda and State House).

Module ownership and back-end delivery

- Registered 30M+ learners across every school level in Uganda by building the national learners module of the Ministry of Education's EMIS platform in Laravel and Vue.js, integrating national identity verification APIs to validate identity records at the point of ingest and eliminate duplicate registrations.
- Delivered structured school-to-location mapping across all administrative units by building the Admin Units module in Laravel with Livewire, Filament, Alpine.js and Tailwind CSS, improving the accuracy and turnaround of national reporting.
- Centralised nationwide staff records by implementing the Laravel HR module with national workforce management, work permit and refugee data integrations, raising data accuracy, regulatory compliance and administrative productivity across the Ministry.
- Enabled long-term, data-driven capital planning by implementing the Infrastructure module to capture school facility data nationwide, directly supporting budgeting for the 2025/2026 financial year and beyond.
- Improved resource allocation and hygiene planning for schools countrywide by developing the Water & Sanitation module to track school utilities, extending the platform's reporting surface without compromising scalability.
- Supported structured uploads of 45M+ contacts segmented by administrative unit by developing the Contact Management module for the Parliament of Uganda BulkSMS platform, improving the precision of targeted government communication.
- Improved support turnaround and accountability across Uganda's education platforms by developing a Laravel ticketing module that logged, routed, tracked and audited system issues and support requests end to end.

- Implemented queue-backed background processing for bulk imports, exports and notification workloads so that long-running jobs never blocked the request cycle or degraded the user experience.

Data modelling, architecture and documentation

- Designed and documented a 1,000+ table PostgreSQL schema, producing the data models, SRS documentation and system architecture that became the reference specification for the EMIS delivery team and measurably improved data integrity and delivery speed.
- Modelled complex real-world relationships — learners, staff, schools, administrative units, facilities and examination records — with constraints, referential integrity and validation enforced at the database layer rather than left to application code.
- Authored technical specifications, interface definitions and architecture decision notes that let new engineers join a large codebase and contribute without a long ramp-up.
- Established migration and versioning discipline for schema change, so that structural updates shipped predictably across development, staging and production.

APIs and system integration

- Streamlined cross-agency data exchange by building secure, scalable APIs in Python FastAPI for national identity (NIRA), examinations board (UNEBC), teacher management (TMIS), Office of the Prime Minister and Ministry of Finance integrations, enhancing the reliability of government service delivery.
- Integrated with undocumented and inconsistent third-party government endpoints, building tolerant clients with retries, timeouts, idempotency keys and reconciliation routines so that partner downtime never corrupted local data.
- Enhanced message reliability, real-time delivery and compliance with strict government security standards by implementing secure SMS integration for the State House BulkSMS platform using Kannel and SMPP servers.
- Secured service-to-service and client access with OAuth 2.0, token-based authentication and role-based authorisation, with audit logging on sensitive operations.

Front-end and user experience

- Delivered data-heavy dashboards, secure API-driven workflows and accessible data-entry screens used daily by ministry staff and district officers nationwide.
- Built complex validated forms and bulk-entry interfaces in Vue.js, Livewire and Filament that reduced user error at source and cut the support load it generated.
- Optimised heavy list and report views with server-side pagination, filtering and query tuning so that screens stayed responsive against tables holding tens of millions of rows.

Quality, performance and release

- Reduced defects reaching production by tightening code review, testing and release checklists, and by verifying data integrity after every deployment.
- Diagnosed and resolved production incidents across application, database and integration layers, establishing root cause and shipping permanent fixes rather than workarounds.
- Maintained containerised environments with Docker and supported Git-based CI/CD release processes, reducing configuration drift between environments.
- Profiled and tuned slow queries and endpoints, using indexing, query rewriting and caching to bring response times back within acceptable limits under national load.

Collaboration and leadership

- Worked directly with ministry stakeholders and district users to elicit requirements, translate them into specifications and manage expectations on scope and timelines.

- Mentored junior engineers through code review, pair debugging and structured feedback, and led technical discussions in sprint planning and team meetings.
- Tracked work in JIRA under Agile/Scrum, contributing to estimation, backlog refinement and release planning.

Eloi Ministries Inc. — Kampala, Uganda (Burlington, USA — Remote) | eloiafrica.org
Software Developer

Feb 2024 – Jul 2024

- Reduced recurring production defects by over 70% by debugging mission-critical software, introducing systematic root-cause analysis and shipping long-term fixes rather than surface patches.
- Improved internal workflows and project delivery by 40% by leading a cross-functional team of software engineers through planning, code review and release, delivering a high-impact project on time and within budget.
- Cut front-end to back-end response times and improved user experience by designing and building a RESTful API layer that replaced ad-hoc data calls across the platform.
- Sustained 99.9% uptime on SQL database servers by owning installation, configuration and ongoing maintenance, guaranteeing uninterrupted access to business-critical application data.
- Launched and maintained two production web platforms (eloiministries.org and eloiministries.us.org), significantly increasing the organisation's digital visibility and outreach.
- Strengthened platform security and compliance by conducting penetration tests with Burp Suite, OWASP ZAP, Nmap and Metasploit, identifying vulnerabilities and driving the remediation work that closed them.
- Reduced post-deployment issues by supporting end-to-end development and QA processes, ensuring application rollouts shipped cleanly and predictably.
- Collaborated across Uganda and United States time zones, working asynchronously with written specifications, clear commit history and documented handovers.
- Coached junior developers on debugging method, testing discipline and readable, maintainable code.

Tuko Super App Technologies Ltd — Kampala, Uganda
Software Engineer

Feb 2023 – Aug 2023

- Supported 100,000+ concurrent users with minimal downtime by designing and building a scalable, high-availability web application architecture, improving user experience and platform engagement.
- Shipped and maintained several production-grade web platforms including tukoapp.com and tukorentals.com using Laravel, Vue.js, Alpine.js and Tailwind CSS, improving reliability, performance and accessibility.
- Hardened platform access by implementing OAuth 2.0 and token-based authentication across services, strengthening data privacy and compliance with security protocols.
- Reduced peak-load query latency and server load by redesigning MySQL and PostgreSQL structures with targeted indexing and query tuning.
- Accelerated development cycles and cross-platform adaptability by building modular codebases spanning PHP (Laravel), JavaScript (Vue.js), Java (Spring Boot) and Python (Flask/Django).
- Accelerated deployment and cross-sector service integration on the Tuko Super App by developing scalable features with Flutter, Spring Boot and Firebase.
- Diagnosed and resolved production incidents affecting live users, communicating status, workaround and resolution clearly to non-technical colleagues.
- Improved release quality by tightening test, review and deployment steps and documenting rollback procedures.

The Voice Mall — Kampala, Uganda
IT Administrator

Mar 2021 – Dec 2023

- Administered the organisation's complete IT infrastructure — physical and virtual servers, workstations, networking equipment and peripherals — ensuring reliable day-to-day operations across the business.
- Set up and maintained local and wireless networks, routers, switches and firewalls, monitoring performance and resolving connectivity issues to minimise downtime.
- Managed user accounts, access permissions and authentication, enforcing role-based access control and least-privilege principles across internal systems.
- Implemented regular data backup and disaster-recovery procedures, safeguarding critical business data against loss and ensuring rapid restoration when needed.
- Enforced IT security best practices — antivirus management, firewall configuration and password policies — protecting the organisation against threats and data breaches.
- Installed, configured and updated operating systems, business software and security patches across all company devices, keeping systems current, stable and secure.
- Provided responsive technical support and troubleshooting to staff, diagnosing and resolving hardware, software and network issues to keep the team productive.
- Documented systems, configurations and support procedures, and managed vendor and ISP relationships through repairs, escalations and service delivery.
- Built the Linux, networking and systems grounding that now underpins deployment, containerisation and production troubleshooting work as an engineer.

SELECTED PROJECTS

EMIS — National Education Management Information System (Government of Uganda) • emis.go.ug

- Contributor on a national platform used across every district in Uganda, spanning Laravel/Vue.js modules, FastAPI integrations and a 1,000+ table PostgreSQL backend managing 30M+ records.
- Integrated the platform with Uganda's national identity registry (NIRA), the Office of the Prime Minister (OPM), the national examinations board (UNEB) and the teacher management information system (TMIS), validating records at ingest and eliminating duplicates across agencies that had never shared data before.
- Delivered data-heavy dashboards, secure API-driven workflows and accessible data-entry screens used daily by ministry staff and district officers nationwide.

Smart Garbage Collection System — Kampala Capital City Authority (KCCA) • impalalite.com

- Contributing to a smart-city waste management platform for KCCA, built ahead of Kampala's smart city plan for 2030, combining IoT hardware data with a microservices backend.
- RFID-based bin scanning to track collection points and monitor fill status across the city, feeding real-time data into the platform for pickup prioritisation.
- Fleet management module covering vehicle tracking, route monitoring and maintenance scheduling for the collection fleet, reducing missed collections and unplanned downtime.
- Built on a microservices architecture with dedicated reporting dashboards, giving KCCA administrators visibility into collection performance, fleet health and city-wide coverage.

BulkSMS Platform — Parliament of Uganda & State House • parliament.smsone.co.ug

- Built contact-management and secure messaging features across a platform handling 45M+ records, with SMPP integration and structured high-volume ingestion.
- Implemented secure delivery through Kannel and SMPP servers under strict government security and reliability standards.

Eloi Ministries Web Platforms • eloiministries.org • eloiafrica.org

- Launched and maintained two production platforms serving a cross-border audience, owning build, deployment, security hardening and ongoing maintenance.

TechTalk — Technology Newsletter (self-initiated)

- Creator and author of a LinkedIn newsletter on technology trends and modern software practices, building an engaged audience through consistent, credible technical writing.

EDUCATION & CERTIFICATION

Diploma in Software Engineering — Sai Pali Institute of Technology & Science, Kampala & Delhi

Sep 2020 – Sep 2022

Relevant coursework: Full-Stack Web Development (HTML, CSS, JavaScript, PHP, Laravel, React), Database Design & Management (MySQL, PostgreSQL, NoSQL), Object-Oriented Analysis & Design (OOAD), Data Structures & Algorithms, Software Architecture & Design Patterns, Agile Methodologies, Scrum & Software Project Management. Completed multiple capstone projects covering the full software lifecycle from planning and architecture through development, testing and maintenance, working in agile teams with Git, Docker and JIRA.

Certificate in Machine Learning — Sandip University, Delhi, India

Jun 2021 – Aug 2022

Supervised and unsupervised learning, neural networks, feature engineering, model evaluation and deployment; Python for machine learning with NumPy, Pandas, Scikit-learn and TensorFlow/Keras.

Continuing development: Ongoing self-directed study in distributed systems and event-driven architecture (Kafka, RabbitMQ), cloud fundamentals (AWS, Azure), observability with the Elastic Stack, and automated testing practice.

PROFESSIONAL STRENGTHS

- Owns problems end to end — from requirements and data model through development, testing, release and production support — rather than handing off at the boundary.
- Writes for the next engineer: clear technical documentation, readable code and specifications that survive handover.
- Communicates clearly with non-technical stakeholders, translating constraints and trade-offs into plain language.
- Effective remotely and asynchronously across time zones, with a disciplined written and commit trail.
- Mentors and reviews constructively, raising the standard of the codebase rather than just the ticket.
- Comfortable in ambiguity — undocumented third-party systems, shifting requirements and legacy data all handled methodically.

ADDITIONAL INFORMATION

Languages: English (fluent) | Luganda (native) | Runyankore-Rukiga (native) | French (basic)

Availability: Immediate. Kampala-based; open to 100% remote engagement (EMEA / GMT+3) or on-site in Uganda.

REFEREES

1. Mr Taqee Ahmed

Senior Software Engineer

Sai Pali Institute of Science and Technology

Tel: +91 8898 333 498 | Email: taqeeahmed@live.com

2. Mr Steven Tendo

Founder

Eloi Ministries Inc. (USA & Uganda) — eloiafrica.org

Email: tendo.steven@gmail.com

3. Mr Clinton

Senior Software Engineer

SMSONE (U) Ltd — smsone.co.ug

Tel: +256 775 599 682